

Targeting tumor-infiltrating myeloid cells to alleviate T cell exhaustion

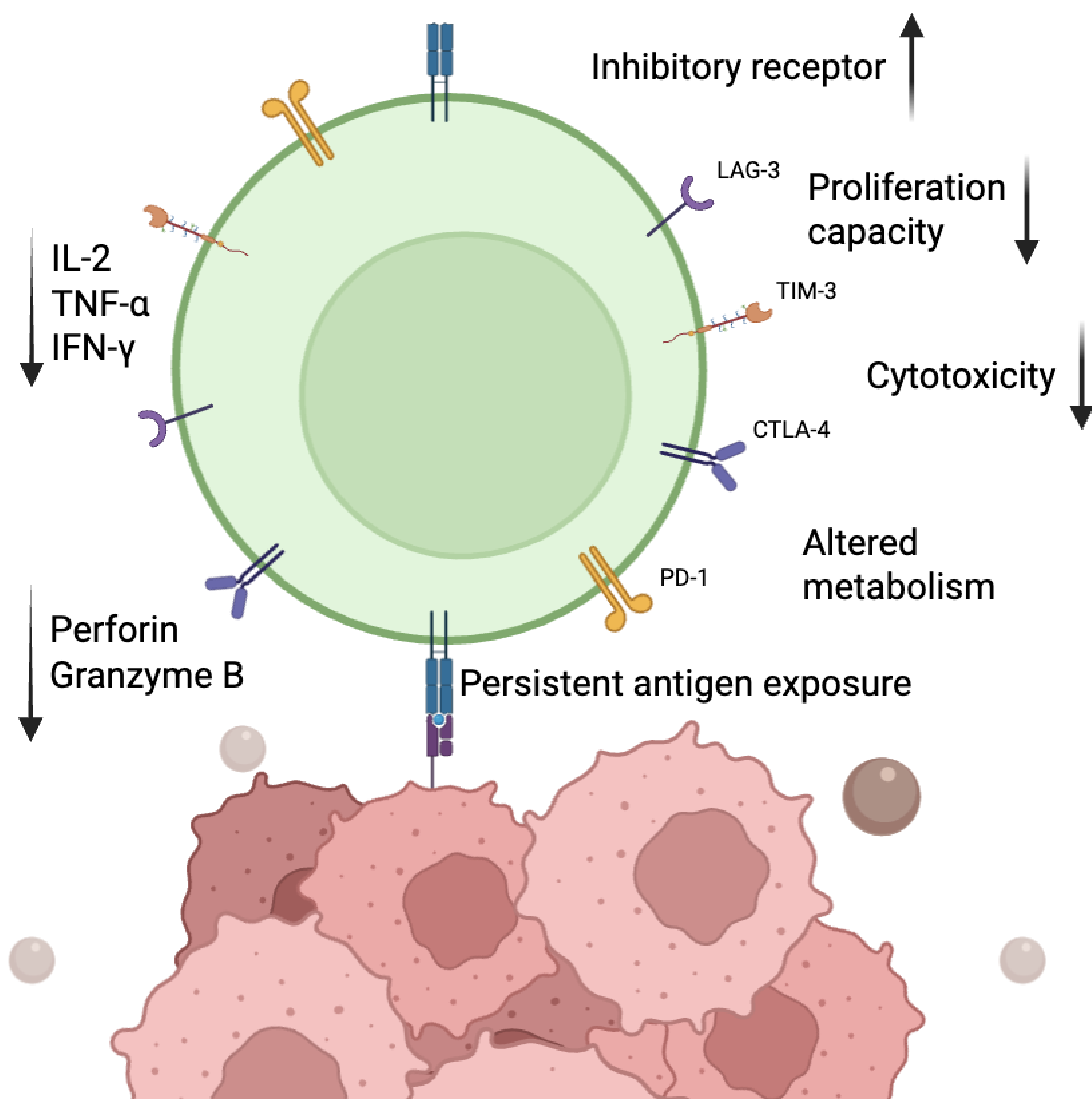
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1. Abstract

T cell exhaustion limits the efficacy of immunotherapies in solid tumors, and tumor-infiltrating myeloid cells (TIMs) are key extrinsic cellular drivers of this dysfunctional state. Understanding how **tumor-associated macrophages** and **myeloid-derived suppressor cells** actively shape T cell exhaustion in the tumor microenvironment (TME) is critical and provides the basis for therapeutic approaches of either **preventing TIM recruitment** or **reprogramming them to anti-tumor phenotypes** to prevent further exhaustion. Major barriers in current treatment mainly include the limited reversible window of exhausted T cells and the functional redundancy of the myeloid population.

2. Introduction to T cell exhaustion

• Loss of effector function

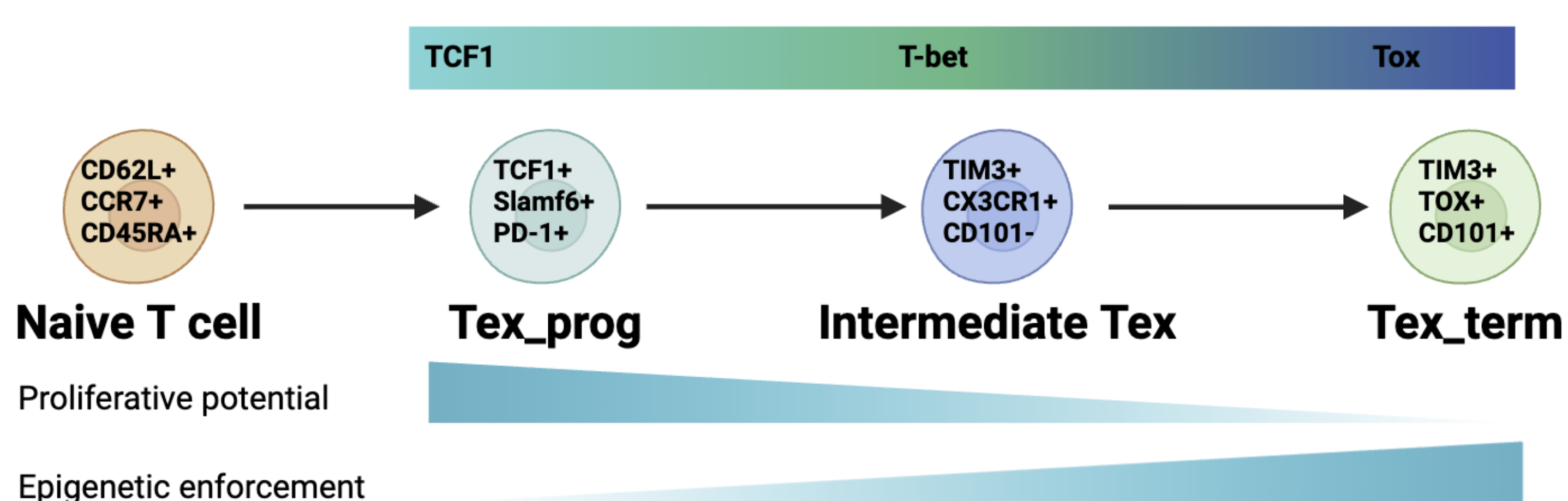


• Transcriptional and epigenetic reprogramming

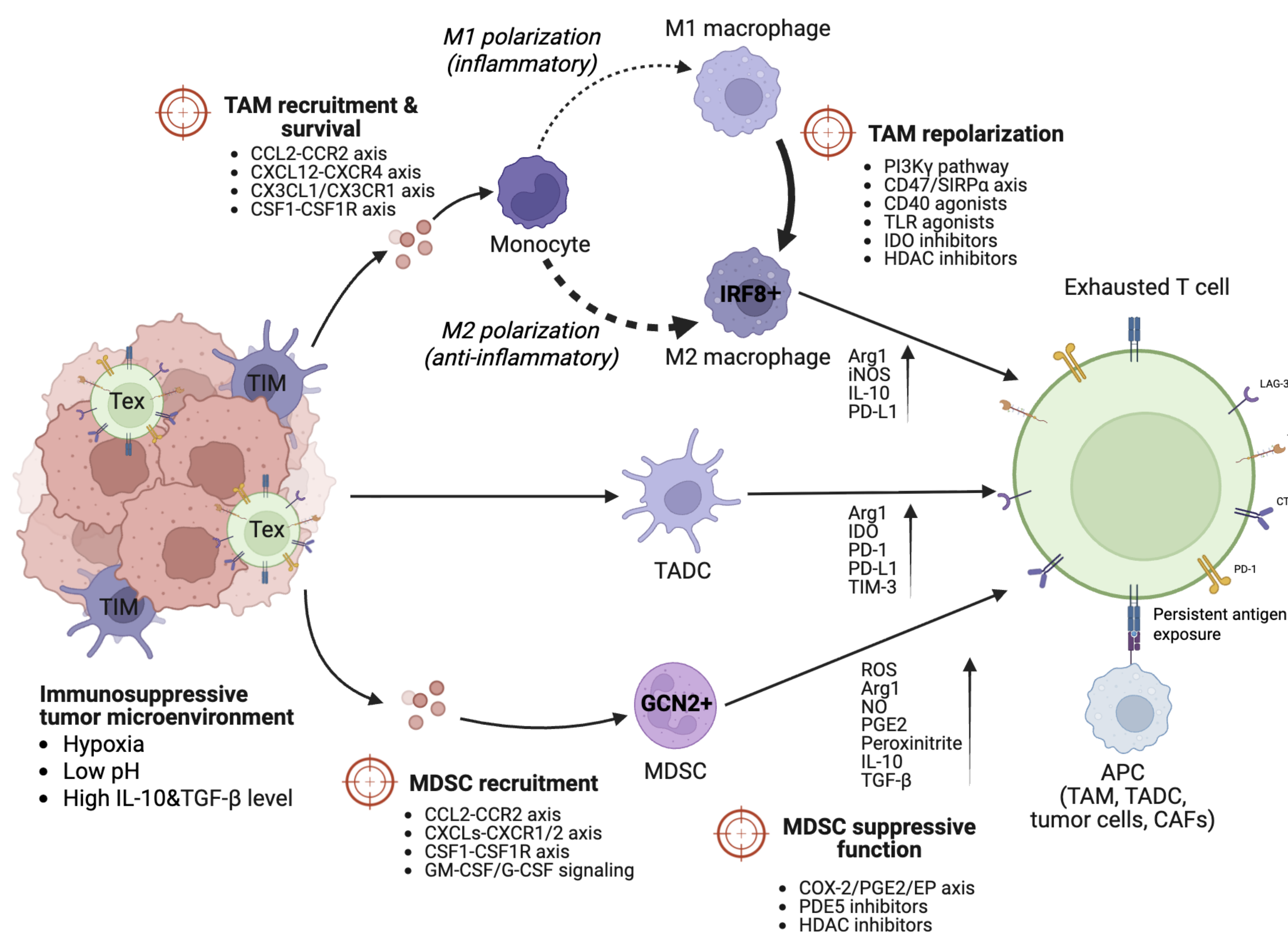
Persistent antigen stimulation reshapes the transcriptional and epigenetic landscape of T cells, locking them into a distinct dysfunctional state. Downstream of chronic TCR signaling, **NFAT** family proteins promote expression of exhaustion-associated genes when lacking their canonical binding partner AP-1. Specifically, unpartnered NFAT induces transcription of *Pdcd1* (PD-1), as well as the master regulator of exhaustion **Tox** and the **Nr4a** family genes (*Nr4a1*, *Nr4a2*, *Nr4a3*).

The epigenetic landscape of Tex is distinct from that of effector (Teff) and memory T cells (Tmem), reflecting **extensive chromatin remodelling and a discrete differentiation lineage**. This is supported by ATAC-seq data, in which 44.48% of chromatin-accessible regions differ between Tex and Teff/Tmem, indicating that exhaustion is not a reversible activation state but an epigenetically fixed program. At the *Pdcd1* locus, an upstream exhaustion-specific enhancer accessible exclusively in Tex, accompanied by selective DNA demethylation at *Pdcd1* and other exhaustion-associated loci, reinforces persistent inhibitory receptor expression.

• Developmental trajectory of Tex



3. Targeting TIMs



• Tumor-associated macrophages (TAMs)

TAMs are the **dominant myeloid population** within the TME and act as a central extrinsic driver of T cell exhaustion. TAMs engage in chronic, low-strength antigen presentation that sustains persistent TCR signaling without productive effector activation. TAMs demonstrate a **spatial-temporal co-dependency with exhausted T cells** in hypoxic, poorly vascularized niches, where metabolic stress further reinforces dysfunction. **IRF8-dependent programming** enables TAMs to acquire a DC-like antigen-presenting phenotype while retaining macrophage identity, promoting PD-1 upregulation and terminal exhaustion through sustained antigen engagement, potentially via peptide-MHC “cross-dressing.” Additionally, TAMs, instead of tumor cells, appear to be the **dominant source of chronic antigen exposure**, driving the transition from progenitor to terminally exhausted T cells.

Therapeutic strategies **targeting TAM recruitment or survival**, including CCL2-CCR2 and CSF1-CSF1R blockade, have shown limited and context-dependent efficacy due to myeloid redundancy and compensatory suppression. An alternative approach is to **repolarize TAM to an anti-tumor phenotype** by blocking pathways such as PI3Kγ and the CD47/SIRPα axis.

• Myeloid-derived suppressor cells (MDSCs)

Myeloid-derived suppressor cells (MDSCs) are a diverse population of aberrantly activated, immature myeloid cells produced during tumor-driven myelopoiesis, where they become **major enforcers** of the immunosuppressive TME. They are broadly categorized into **monocytic MDSCs (M-MDSCs)** and **granulocytic/polymorphonuclear MDSCs (PMN-MDSCs)**, each deploying distinct suppressive programs once within the tumor. In the TME, M-MDSCs produce **high levels of nitric oxide, IL-10, TGF-β, and upregulate PD-L1**, whereas PMN-MDSCs generate **reactive oxygen species, peroxynitrite, arginase-1, and prostaglandin E2 (PGE2)**.

The major therapeutic strategies focus on **targeting the recruitment of MDSCs** and **blocking their immunosuppressive effector function**.

4. Current challenges

- Irreversibility of terminal exhaustion
- Persistent antigen exposure is not eliminated
- Myeloid redundancy and compensatory suppression
- Overlooked CD4⁺ T cell exhaustion

